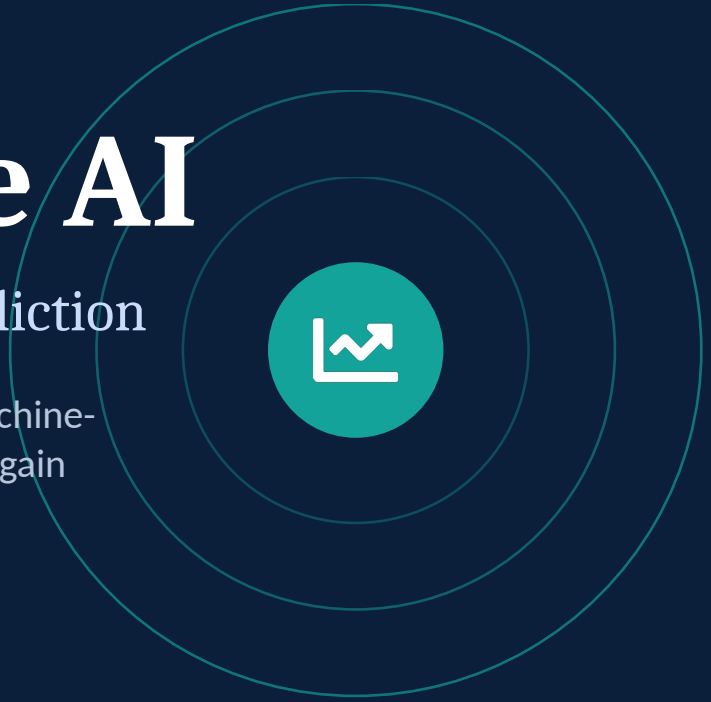
RETAIL ANALYTICS · PORTFOLIO CASE STUDY

CommercePulse AI

Retail analytics & repeat-purchase prediction



A Power BI retail intelligence dashboard paired with a machine-learning model that estimates which customers will buy again within three months.

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GitHub · CommercePulse-AI

From transaction records to decisions

PROBLEM STATEMENT

Retailers hold thousands of invoices, but raw transaction logs do not show which products, markets and customers drive performance — or who is likely to return.

CommercePulse AI turns that data into a view of what happened and an estimate of what may happen next.

PROJECT OBJECTIVES



Sales performance

How much is sold, and when do sales peak?



Product contribution

Which products generate the most sales value?



Market mix

Which countries carry the business?



Customer behaviour

Who is likely to purchase again in 3 months?

Why it matters: these answers guide stock planning, product focus, market priorities and retention effort.

From raw invoices to a predictive model

14.84M	30K	8M	43
Sales value before returns	Invoices	Units sold	Countries

Online retail transactions · 1 Dec 2009 – Jul 2011 (date range in dashboard filter)

1 Raw transactions
Invoice-level retail records

2 Inspect & audit
Check structure and data quality

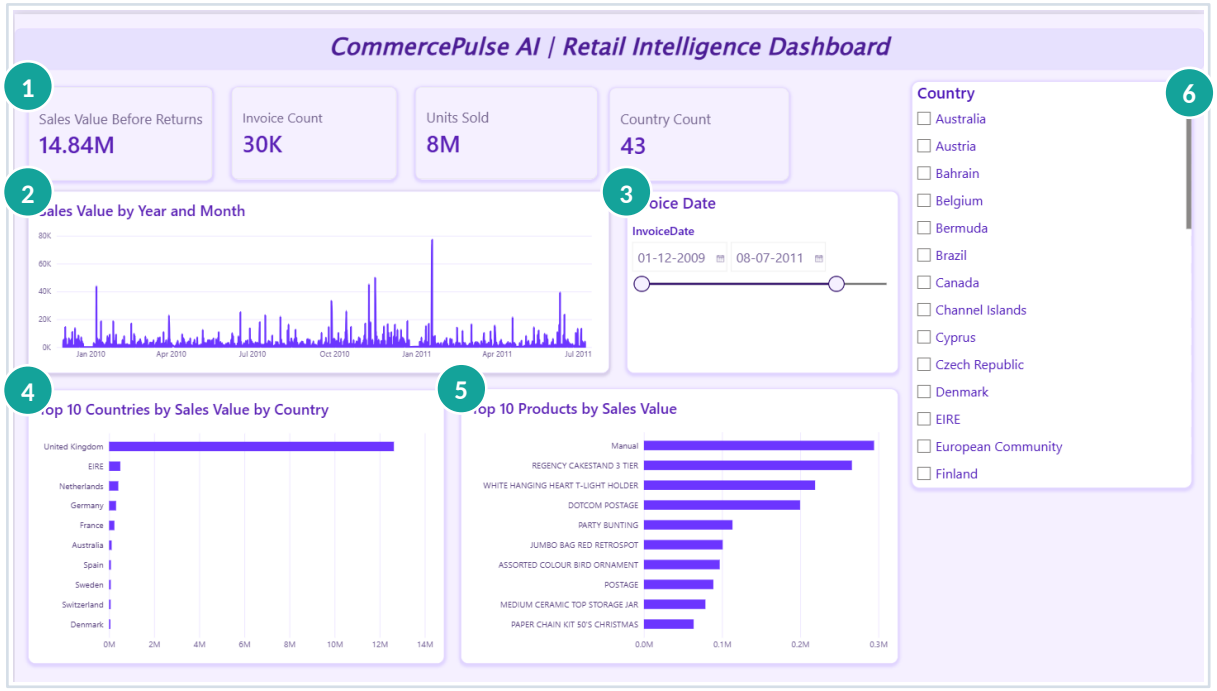
3 Prepare
Shape data for analysis

4 Business analysis
Sales, products, countries, customers

5 Power BI dashboard
KPIs, trends and filters

6 Predictive model
Random Forest vs Logistic Regression

Retail Intelligence Dashboard (Power BI)



1. **KPI cards**
   14.84M sales before returns, 30K invoices, 8M units, 43 countries
2. **Sales trend**
   Sales value over time, Jan 2010 – Jul 2011
3. **Date filter**
   Slider and date boxes narrow the period
4. **Top 10 countries**
   Ranks markets by sales value
5. **Top 10 products**
   Ranks products by sales value
6. **Country slicer**
   Tick countries to filter the page

Three things the dashboard reveals

≈85%

of sales value comes from the United Kingdom

EVIDENCE

About 12.6M of the 14.84M total. EIRE, the Netherlands and Germany follow far behind.

SO WHAT

Market strategy hinges on one country; growth elsewhere starts from a small base.

3 of 10

top 'products' are not merchandise

EVIDENCE

Manual, Dotcom Postage and Postage sit in the Top 10 by sales value.

SO WHAT

Separate service and shipping lines from merchandise before ranking products for stocking.

≈78K

largest single sales peak, near Jan 2011

EVIDENCE

Sales run low most days, broken by sharp spikes of roughly 40K or more.

SO WHAT

Planning on averages would hide these peaks; inventory should be tested against them.

Figures are read from the Power BI dashboard; values taken from chart axes are approximate.

05 REPEAT-PURCHASE PREDICTION

Will this customer buy again within 3 months?

Past purchases (features) ⟩ **Next 3 months (repeat purchase?)**

Temporal backtest: train on earlier history, evaluate on later periods the model has not seen.

MODELS COMPARED

- **Random Forest**
- **Logistic Regression**

BACKTEST METRICS

Metric	Random Forest	Logistic Reg.
ROC-AUC	*[insert]*	*[insert]*
Precision	*[insert]*	*[insert]*
Recall	*[insert]*	*[insert]*
F1	*[insert]*	*[insert]*

Selected model: [Random Forest / Logistic Regression] — [one-line reason, e.g. best score on the key metric]

Historical backtest only. The model is not deployed and no campaign results are claimed.

KEY TAKEAWAYS

Describe the past, estimate the next step

Descriptive

Power BI dashboard on 14.84M of sales across 43 countries, with filters for date and country.

Predictive

Repeat-purchase model with a 3-month horizon, tested on historical time splits against two algorithms.

Integrated

One workflow from raw transactions to dashboard to model, so each stage informs the next.

Potential uses: inventory planning, product prioritisation, market analysis and retention targeting — each still to be validated in practice.

Good decisions need both a clear view of what happened and a measured estimate of what comes next.

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